

Day 11 - Reverse a Number ☐

Question

Ek integer N diya gaya hai.

Uske digits ko reverse karke print karo.

Example 1

Input

12345

Output

54321

Example 2

Input

500

Output

5

Leading zeros automatically remove ho jate hain.

Example 3

Input

-123

Output

-321

Approach 1 - Mathematical Method

(Interview Best)

Logic

Har step me:

1. Last digit nikalo.
2. Reverse number me add karo.
3. Original number se last digit hata do.

Formula

```
digit = n % 10
```

```
reverse = reverse * 10 + digit
```

```
n = n / 10
```

Dry Run

Input

```
n = 1234
```

```
n  digit = n%10 reverse
```

```
1234 4      4
```

```
123  3      43
```

```
12   2      432
```

```
1    1      4321
```

```
0    -      Stop
```

Output:

```
4321
```

Java Code

```
public class Day11 {  
    public static void main(String[] args) {  
        int n = 1234;  
        int reverse = 0;  
        while(n != 0) {  
            int digit = n % 10;  
            reverse = reverse * 10 + digit;  
            n = n / 10;  
        }  
        System.out.println(reverse);  
    }  
}
```

Time Complexity

$O(\log_{10} N)$

Reason:

Digits jitne honge utni hi iterations chalengi.

Space Complexity

$O(1)$

Approach 2 - Using StringBuilder

Logic

Number ko String me convert karo.

Reverse karo.

Phir Integer me convert karo.

Java Code

```
public class Day11 {  
    public static void main(String[] args) {  
        int n = 1234;  
        String str = Integer.toString(n);  
        StringBuilder sb = new StringBuilder(str);  
        sb.reverse();  
        System.out.println(sb);  
    }  
}
```

Time Complexity

$O(N)$

(N = Number of digits)

Space Complexity

$O(N)$

Extra String memory use hoti hai.

Approach 3 - Using Recursion

Logic

Har recursive call me ek digit process hoti hai.

Java Code

```
public class Day11 {  
    static void reverse(int n) {  
        if(n == 0)  
            return;  
        System.out.print(n % 10);  
        reverse(n / 10);  
    }  
    public static void main(String[] args) {  
        reverse(1234);  
    }  
}
```

Output

4321

Time Complexity

$O(\log N)$

Space Complexity

$O(\log N)$

(Call Stack)

Approach 4 - Handling Negative Numbers



Java Code

```
public class Day11 {  
    public static void main(String[] args) {  
        int n = -123;  
        boolean negative = n < 0;  
        if(negative)  
            n = -n;  
        int reverse = 0;  
        while(n != 0) {  
            reverse = reverse * 10 + (n % 10);  
            n /= 10;  
        }  
        if(negative)  
            reverse = -reverse;  
        System.out.println(reverse);  
    }  
}
```

Output

-321

Overflow Problem ☐

Suppose:

`n = 2147483647`

Reverse:

`7463847412`

Ye `int` me fit nahi hota.

Isliye interview me overflow check karna important hai.

Safe Code

```
if (reverse > Integer.MAX_VALUE / 10 ||  
    reverse < Integer.MIN_VALUE / 10) {  
  
    System.out.println("Overflow");  
    return;  
}
```

Interview Follow-Up Questions

Q1. Why use $\% 10$?

$$123 \% 10 = 3$$

Last digit mil jati hai.

Q2. Why use $/ 10$?

$$123 / 10 = 12$$

Last digit remove ho jati hai.

Q3. Why multiply reverse by 10?

Example:

$$\text{reverse} = 43$$

$$\text{New digit} = 2$$

$$43 \times 10 + 2$$

$$= 432$$

New digit ko last me add karne ke liye.

Interview Concept of the Day

Number Digit Extraction

Ye 2 formulas bahut important hain.

```
digit = n % 10;
```

Last digit nikalta hai.

```
n = n / 10;
```

Last digit remove karta hai.

Future me ye same logic use hoga:

- Palindrome Number
- Armstrong Number
- Sum of Digits
- Count Digits
- Happy Number
- Strong Number

Summary

Approach	Time	Space
Mathematical Method	$O(\log N)$	$O(1)$
StringBuilder	$O(\text{Digits})$	$O(\text{Digits})$
Recursion	$O(\log N)$	$O(\log N)$
Negative Handling	$O(\log N)$	$O(1)$

Most Asked Interview Test Cases

Input Output

1234 4321

500 5

1000 1

-123 -321

0 0

Practice Questions

Reverse these numbers:

123
450
7890
90001
-5678

Interview Tip ☐

Agar interviewer bole:

"Reverse a number without using String."

To **Mathematical Method** ($\% 10$ aur $/ 10$) use karo. Ye expected aur optimal solution hai.